Video-RAG: Visually-Aligned Retrieval-Augmented Long Video Comprehension

Anonymous Authors

Abstract

Long video comprehension remains difficult for large video-language models (LVLMs) because a fixed context budget must cover both long-range temporal evidence and fine-grained visual details. A common response is to feed more frames or to rely on multi-stage agent systems, but both choices are expensive and often require long-context training or proprietary models. We present *Video-RAG*, a training-free retrieval-augmented pipeline that improves long-video question answering by replacing part of the redundant visual context with query-relevant auxiliary text extracted from the same video. Given a question, Video-RAG first asks the LVLM itself to emit a structured retrieval request, then retrieves supporting evidence from three video-specific sources: automatic speech recognition transcripts, optical character recognition strings, and open-vocabulary object detections converted into scene-graph descriptions. The retrieved texts are concatenated ahead of the final question in a single turn, preserving the original LVLM and its frame budget. Across seven open-source LVLMs on Video-MME, Video-RAG improves over subtitle-assisted baselines by 2.8 points on average while adding only about 2.0K auxiliary-text tokens per sample. LLaVA-Video-72B with Video-RAG reaches 77.4% on Video-MME, surpassing GPT-4o at 77.2%. The same plug-and-play strategy also improves LLaVA-Video on MLVU from 70.8% to 72.4% at 7B and from 73.1% to 73.8% at 72B, and raises LongVideoBench validation accuracy from 61.9% to 65.4% at 72B. These results show that query-aligned auxiliary text is an effective way to improve long-video reasoning without model retraining.

1 Introduction

Large video-language models (LVLMs) have rapidly improved short-video understanding, but long-video comprehension remains a bottleneck. The core difficulty is not only temporal length. Long videos mix sparse query-relevant evidence with large amounts of redundant content, so simply sampling more frames often wastes the context budget on irrelevant visual tokens. This problem is especially visible on recent long-video benchmarks such as Video-MME, MLVU, and LongVideoBench, where performance degrades as duration grows and success depends on retrieving the right evidence rather than processing every frame uniformly [4, 37, 30].

Existing solutions usually follow one of two directions. The first extends the visual context window through architectural changes, visual compression, or long-context transfer [25, 34, 22, 24, 15]. These methods can be effective, but they still pay for additional visual tokens and often depend on long-context adaptation. The second direction delegates reasoning to multi-stage agents or external planners that repeatedly call tools and proprietary language models [29, 16, 33]. Such systems improve interpretability and retrieval, but they are operationally heavier and less convenient as a drop-in upgrade for an arbitrary open-source LVLM.

Our starting point is a different observation: long videos already contain a large amount of useful text-like evidence that can be extracted cheaply from the video itself. Speech transcripts, overlaid

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

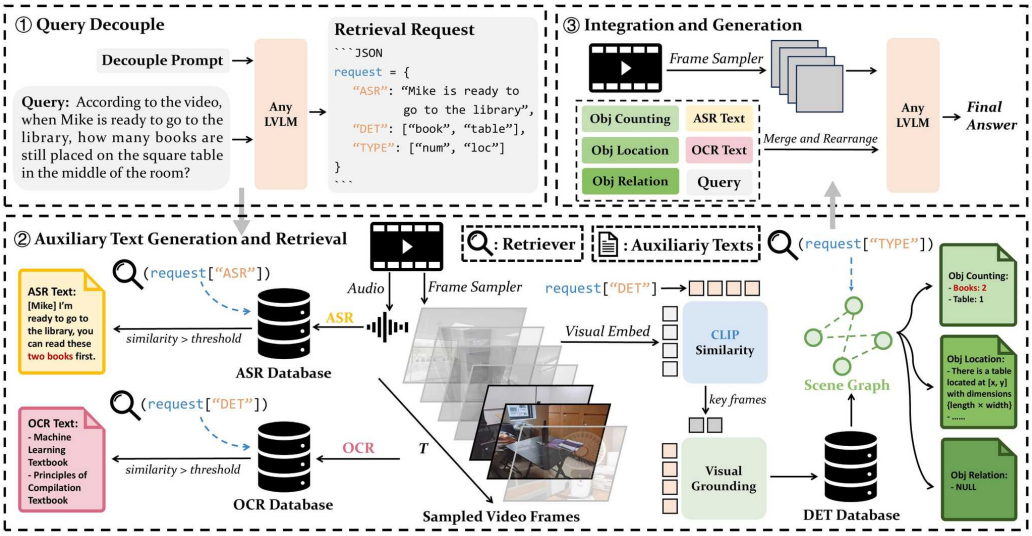

Figure 1: Overview of Video-RAG. The method first decouples the question into a structured retrieval request, then retrieves auxiliary evidence from video-specific OCR, ASR, and object sources, and finally prepends the retrieved texts to the original question before a standard LVLM forward pass. OCR and ASR retrieval are text-based, while object retrieval first selects keyframes with CLIP similarity and then converts APE detections into scene-graph descriptions.

text, and object-level visual descriptions are often more compact than dense frame tokens and more directly aligned with the reasoning needs of multiple-choice video QA. If these signals are retrieved in a query-aware way, they can serve as targeted memory that steers the LVLM toward relevant frames and away from distractors.

Based on this observation, we propose *Video-RAG*, a training-free and plug-and-play retrieval-augmented generation pipeline for long-video comprehension. Video-RAG first asks the LVLM to decompose the user query into a small JSON retrieval request specifying which speech content, physical entities, and object attributes should be retrieved. It then gathers evidence from three sources built from the same video: automatic speech recognition (ASR), optical character recognition (OCR), and open-vocabulary detections. ASR and OCR texts are retrieved with Contriever [6] and FAISS [8], while object requests use CLIP [18] to identify relevant frames and APE [23] to produce detections that are converted into scene-graph descriptions. The selected evidence is inserted before the final question in a single LVLM call, without changing model weights.

This design leads to a practical trade-off. Instead of scaling the number of visual tokens, Video-RAG injects a small amount of query-relevant text into the existing prompt. On Video-MME, the method improves seven open-source LVLMs by 2.8 points on average relative to subtitle-assisted baselines. On LLaVA-Video-72B [36], it reaches 77.4%, exceeding GPT-4o [17] in the same benchmark setting. The same recipe also improves MLVU and LongVideoBench, with particularly strong gains on long videos and long-context models.

Our contributions are threefold:

- We introduce a training-free retrieval pipeline for long-video QA that augments an LVLM with query-relevant OCR, ASR, and object-level auxiliary text extracted from the input video itself.
- We show that visually aligned retrieval, especially object-derived scene-graph text, improves cross-modal grounding without increasing the number of video frames.
- We demonstrate consistent gains on Video-MME, MLVU, and LongVideoBench across multiple open-source LVLMs, including proprietary-level performance with an open 72B model.

Figure 2 summarizes the intended operating point of our method. Relative to long-context LVLMs, Video-RAG does not require extending the visual context window. Relative to agent pipelines, it avoids repeated reasoning loops and external proprietary backends. The result is a middle ground that remains training-free while still performing explicit evidence selection.

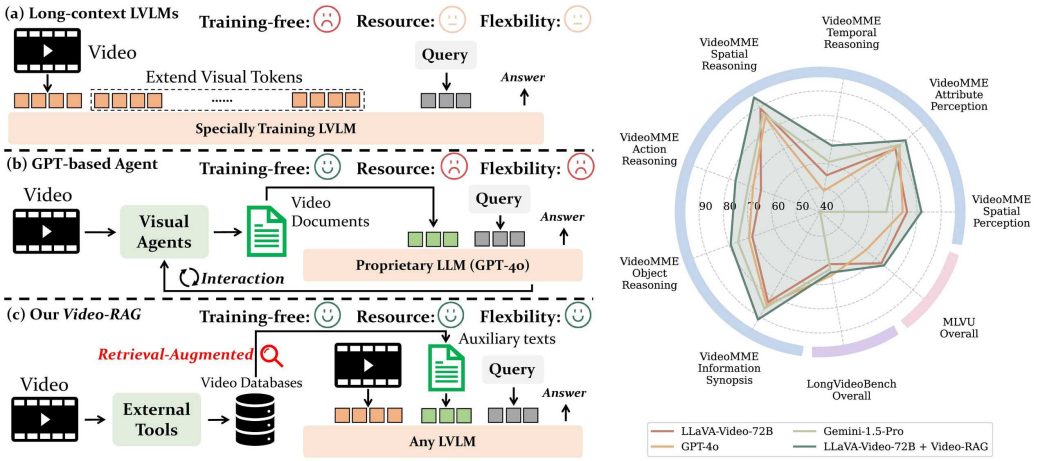

Figure 2: Motivation and high-level comparison. Long-context LVLMs improve coverage by processing more visual tokens, while agent systems convert video content into external documents through multi-stage interaction. Video-RAG instead retrieves compact auxiliary text from the queried video and keeps the final reasoning step within a standard LVLM call. The radar chart summarizes that this strategy improves multiple Video-MME dimensions and also transfers to MLVU and LongVideoBench.

2 Related Work

Long-video LVLMs. Many recent methods attack long-video understanding by expanding the amount of visual evidence the model can ingest. LongVA transfers long-context ability from language backbones to vision [34], while INTP-Video-LLMs extends sequence handling at test time without additional video training [22]. Other methods introduce memory or compression modules, such as MovieChat's sparse memory [25], Video-XL's summarization tokens [24], and Oryx's dynamic visual compression [15]. These approaches improve coverage, but their main lever is still visual-token management. Video-RAG is complementary: instead of consuming more visual tokens, it inserts compact auxiliary text retrieved from the video and can therefore benefit existing LVLMs without modifying their architecture.

Agent and modular video reasoning. Another line of work decomposes video QA into tool use, planning, or modular reasoning. VideoAgent treats the language model as an agent that iteratively gathers evidence [29], and MoReVQA decomposes video QA into event parsing, grounding, and reasoning stages [16]. OmAgent similarly emphasizes divide-and-conquer strategies for complex video understanding [33]. These systems are effective for complex reasoning, but they usually require multi-stage interaction and may depend on larger external LLMs. Video-RAG keeps the interaction single-turn: the model emits one retrieval request, the three retrieval branches run in parallel, and the augmented prompt is answered by the same LVLM.

Textualization and retrieval for video understanding. Several works rely on converting video into text before language reasoning. LLoVi uses dense short-term captions followed by long-range textual aggregation [32], and earlier systems such as VLog also treat videos as text-heavy documents [13]. Retrieval-augmented generation has been widely successful in language [9] and has also been adapted to video corpora [1]. Our setting is different from corpus-level video retrieval: we retrieve within a single queried video and use the retrieved text to reallocate the prompt budget away from redundant frames and toward query-specific evidence. This makes retrieval a mechanism for *intra-video alignment*, not only for inter-video search.

3 Method

3.1 Problem Setting

Given a video V, a multiple-choice question q, and an LVLM f, the goal is to answer the question using a fixed visual budget of sampled frames. A vanilla LVLM receives uniformly sampled frames and the question, but performance drops on long videos because many sampled frames are weakly related to the question. Video-RAG augments the same inference-time pipeline with retrieved auxiliary text $\mathcal{A}$ extracted from V, and answers with $f(V, \mathcal{A}, q)$ in a single forward pass.

3.2 Overview

Figure 1 shows the pipeline. Video-RAG has three stages:

1. **Query decoupling.** The LVLM receives only the question and outputs a JSON retrieval request with three fields: an ASR hint, a list of physical entities for object retrieval, and a `TYPE` field indicating whether location, counting, or spatial relation information is needed.
2. **Auxiliary-text generation and retrieval.** We build OCR, ASR, and detection evidence from the video, then retrieve only the query-relevant subset.
3. **Integration and generation.** The selected texts are inserted before the original question in DET→ASR→OCR order and passed, together with the sampled frames, to the LVLM for answer generation.

All stages are training-free. The only learned components are off-the-shelf open-source models used as frozen tools.

3.3 Query Decoupling

The decoupling prompt asks the LVLM to produce a structured request rather than an answer. Concretely, it predicts:

- `ASR`: a short textual description of the speech content worth retrieving;
- `DET`: up to a few physical entities relevant to the question;
- `TYPE`: whether the model needs object `location`, `number`, and/or `relation`.

This step is inexpensive because it does not consume video tokens. It also keeps retrieval aligned with the answering model: the same LVLM that will answer the question specifies which evidence it wants.

3.4 Building Video-Specific Auxiliary Corpora

ASR corpus. We extract the video audio with `ffmpeg` and transcribe it using Whisper-large [19]. The implementation chunks audio into 30-second segments and stores each transcript chunk as an ASR document.

OCR corpus. We sample video frames and run EasyOCR [7]. OCR strings with confidence above 0.5 are kept, duplicates across frames are removed, and the remaining strings are stored in chronological order as OCR documents.

Detection corpus. Object evidence is built in two steps. First, we score each sampled frame against the requested object phrases using CLIP-ViT-L/14-336 [18]. Let v_i be the feature of frame i and c_k the feature of requested object phrase k. We compute the average object relevance

$$\bar{s}_i = \frac{1}{K} \sum_{k=1}^{K} \langle v_i, c_k \rangle, \tag{1}$$

then rescale it exactly as in the implementation,

$$s_i = \frac{\alpha \bar{s}_i}{\sum_{u=1}^{T} \bar{s}_u}, \qquad \alpha = \beta T/16, \tag{2}$$

where T is the number of sampled frames and $\beta = 3.0$. Frames with $s_i > 0.3$ are sent to APE [23], an open-vocabulary detector configured with score threshold 0.1. APE returns bounding boxes for the requested object names, which we convert into scene-graph text describing object counts, coordinates, and pairwise spatial relations depending on the requested `TYPE`.

3.5 Text Retrieval

OCR and ASR retrieval use Contriever [6] with FAISS inner-product search [8]. If the request produces a set of query strings $\{q_1, \ldots, q_m\}$, we encode each query and average the embeddings:

$$\mathbf{q} = \mathrm{norm}\left(\frac{1}{m}\sum_{i=1}^{m} E(q_i)\right), \tag{3}$$

where $E(\cdot)$ is the mean-pooled Contriever encoder. For a document set $\mathcal{D} = \{d_j\}$, we retrieve

$$\mathcal{R} = \{d_j \in \mathcal{D} \mid \langle \mathbf{q}, \mathrm{norm}(E(d_j)) \rangle > t\}, \tag{4}$$

using `faiss.IndexFlatIP` range search with threshold $t = 0.3$. OCR retrieval uses the original question plus requested object keywords. ASR retrieval uses the question plus the optional ASR hint. The three retrieval branches are independent and can run in parallel.

3.6 Prompt Integration

The final prompt concatenates retrieved detection descriptions, ASR snippets, and OCR snippets before the answer instruction. This ordering is fixed in the code: detection text first, ASR second, OCR third. The key effect is to provide a compact textual scaffold that highlights where relevant evidence lies before the LVLM reasons over the sampled frames. In practice, this often strengthens alignment between the question and a small number of relevant keyframes instead of forcing the model to attend uniformly across all frames.

4 Experiments

4.1 Benchmarks and Setup

We evaluate on three long-video multiple-choice benchmarks.

- **Video-MME** [4]: a large-scale benchmark spanning short, medium, and long videos with settings both without subtitles and with subtitles.
- **MLVU** [37]: a long-video benchmark with multiple tasks; we report overall accuracy on the multiple-choice split.
- **LongVideoBench** [30]: a long-context interleaved video-language benchmark; we report validation accuracy.

Video-RAG is evaluated in a zero-shot, training-free setting. For Video-MME, we follow the native frame budgets of the compared LVLMs shown in Table 1. For MLVU and LongVideoBench, we evaluate LLaVA-Video [36] at 7B and 72B with 64 sampled frames. The retrieval stack uses EasyOCR [7], Whisper-large [19], CLIP-ViT-L/14-336 [18], APE [23], Contriever [6], and FAISS [8]. Unless stated otherwise, the textual retrieval threshold is $t = 0.3$ and the CLIP frame-selection threshold is also 0.3.

In the implementation, frames are sampled uniformly up to the model-specific budget, and the same answering LVLM is also used for the query-decoupling step without visual input. OCR strings are deduplicated across frames, ASR is chunked into 30-second segments, and the detector request is restricted to a small set of concrete entities. All benchmark results are obtained without task-specific fine-tuning; Video-RAG only changes test-time retrieval and prompt construction.

4.2 Main Results on Video-MME

Table 1 shows that Video-RAG consistently improves all seven open-source LVLMs over subtitle-assisted baselines. The average improvement is 2.8 points, and gains are particularly pronounced for

Table 1: Performance on Video-MME [4] without subtitles (w/o S), with subtitles (w/ S), and with Video-RAG (Ours). **Frames** denotes the number of sampled frames and **Gain** is the improvement of Video-RAG over the subtitle-assisted baseline. Across seven open-source LVLMs, Video-RAG improves accuracy by 2.8 points on average while adding only ∼2.0K auxiliary-text tokens per sample, compared with ∼3.0K subtitle tokens. On LLaVA-Video-72B [36], Video-RAG reaches 77.4%, slightly above GPT-4o [17] at 77.2%. Results are reproduced in our evaluation setup.

Model	Params	Frames	Short			Medium			Long			Overall			Gain
			w/o S	w/ S	Ours	w/o S	w/ S	Ours	w/o S	w/ S	Ours	w/o S	w/ S	Ours	
Proprietary LVLMs															
GPT-4o [17]	-	384	80.0	82.8	-	70.3	76.6	-	65.3	72.1	-	71.9	77.2	-	-
Gemini-1.5-Pro [20]	-	0.5 fps	81.7	84.5	-	74.3	81.0	-	67.4	77.4	-	75.0	81.3	-	-
Open-Source LVLMs															
Video-LLaVA [12]	7B	8	45.3	46.1	49.5	38.0	40.7	43.0	36.2	38.1	42.5	39.9	41.6	45.0	+3.4
LLaVA-NeXT-Video [35]	7B	16	49.4	51.8	56.6	43.0	46.4	47.4	36.7	44.9	46.0	43.0	47.7	50.0	+2.3
VITA-1.5 [5]	7B	16	67.0	69.9	71.0	54.2	55.7	55.4	47.1	50.4	52.4	56.1	58.7	59.6	+0.9
LongVA [34]	7B	128	61.1	61.2	66.1	50.4	53.8	60.4	46.2	52.9	59.4	52.6	56.0	62.0	+6.0
Long-LLaVA [26]	7B	64	61.9	62.4	67.1	51.4	56.2	60.4	45.4	54.7	60.1	52.9	57.8	62.6	+4.8
Qwen2-VL [28]	72B	32	75.0	76.7	77.4	63.3	69.9	70.2	56.3	69.2	71.0	64.9	71.9	72.9	+1.0
LLaVA-Video [36]	72B	64	80.7	81.8	82.8	68.7	73.8	76.3	62.1	72.2	73.1	70.3	75.9	77.4	+1.5

7B long-context models. LongVA [34] improves from 56.0% to 62.0% overall, while Long-LLaVA [26] improves from 57.8% to 62.6%. The long-video slice benefits the most for these models: LongVA rises from 52.9% to 59.4%, and Long-LLaVA rises from 54.7% to 60.1%.

The strongest absolute result comes from LLaVA-Video-72B [36], which improves from 75.9% with subtitles to 77.4% with Video-RAG. This slightly exceeds GPT-4o at 77.2% in the same benchmark table and narrows the gap to Gemini-1.5-Pro at 81.3%. Importantly, these gains come from relatively compact auxiliary text instead of dense subtitle context, supporting the claim that query-aware evidence is more useful than simply appending more raw text.

The comparison against subtitle baselines is important. Subtitles already provide a strong textual cue on Video-MME, but they are not always well aligned with the question and they omit visual facts such as object counts, positions, and scene text. Video-RAG addresses this mismatch by retrieving only the text that is relevant to the question and by adding visual descriptions that subtitles do not contain. This explains why the method improves not only models with weak long-context ability, such as Video-LLaVA and LLaVA-NeXT-Video, but also stronger long-context models such as LongVA and Long-LLaVA.

4.3 Generalization to MLVU and LongVideoBench

Table 2: Overall accuracy on the multiple-choice task of MLVU [37]. * denotes reproduced baseline results in our setup.

Model	#Params	Frames	Overall
Proprietary LVLMs			
GPT-4o [17]	-	0.5 fps	64.6
Open-Source LVLMs			
VITA-1.5 [5]	7B	16	60.4
Video-CCAM [3]	14B	96	63.1
Video-XL [24]	7B	256	64.9
Aria [10]	25.3B	256	70.6
LLaVA-Video* [36]	7B	64	70.8
Oryx-1.5 [15]	32B	128	72.3
LLaVA-Video* [36]	72B	64	73.1
LLaVA-Video + Video-RAG	7B	64	72.4
LLaVA-Video + Video-RAG	72B	64	**73.8**

Table 3: Overall accuracy on the validation set of LongVideoBench [30]. * denotes reproduced baseline results in our setup.

Model	#Params	Frames	Overall
Proprietary LVLMs			
Gemini-1.5-Pro [20]	-	256	64.0
GPT-4o [17]	-	256	**66.7**
Open-Source LVLMs			
VideoChat2-Mistral [11]	7B	8	39.3
ShareGPT4Video [2]	7B	8	39.7
LLaVA-Next-Mistral [14]	7B	8	49.1
PLLaVA [31]	34B	16	53.2
VITA-1.5 [5]	7B	16	53.6
LLaVA-Video* [36]	7B	64	56.6
LLaVA-Video* [36]	72B	64	61.9
LLaVA-Video + Video-RAG	7B	64	58.7
LLaVA-Video + Video-RAG	72B	64	65.4

Table 2 and Table 3 show that the same retrieval pipeline transfers beyond Video-MME. On MLVU, LLaVA-Video-7B improves from 70.8% to 72.4%, and LLaVA-Video-72B improves from 73.1% to

73.8%, the best score among the compared methods. On LongVideoBench validation, LLaVA-Video-72B improves from 61.9% to 65.4%, surpassing Gemini-1.5-Pro at 64.0% and approaching GPT-4o at 66.7%. The 7B model also improves from 56.6% to 58.7%.

These results matter because the three benchmarks stress different failure modes. Video-MME emphasizes broad multimodal understanding [4], MLVU targets long-video reasoning across diverse tasks [37], and LongVideoBench stresses referred-context retrieval in long interleaved inputs [30]. The consistent gains across all three suggest that Video-RAG is not tied to a specific benchmark annotation style.

5 Analysis

5.1 Which Auxiliary Text Matters?

Table 4: Ablation on auxiliary-text design on Video-MME [4] with Long-LLaVA-7B [26]. **RAG** indicates whether query-aware retrieval is enabled before concatenating the selected texts.

RAG	DET	OCR	ASR	Short	Medium	Long	Overall
				60.3	51.4	44.1	52.0
			✓	62.2	55.4	54.4	57.4
		✓	✓	64.0	56.2	55.0	58.4
	✓		✓	63.0	57.3	**56.4**	58.9
	✓	✓	✓	**64.3**	**58.8**	56.3	**59.8**
✓	✓			61.4	51.9	45.2	52.9
✓		✓		63.2	53.2	46.3	54.3
✓			✓	65.1	59.1	**60.7**	61.6
✓	✓	✓		64.1	54.6	48.4	55.7
✓	✓		✓	64.9	59.0	**60.7**	61.5
✓		✓	✓	66.3	**60.3**	59.3	62.0
✓	✓	✓	✓	**66.4**	60.2	59.8	**62.1**

Table 5: Effect of the retrieval threshold on Video-MME [4] with Long-LLaVA-7B [26]. **#Token** and **Time** denote the total auxiliary-text token count and the average inference time per question, respectively.

t	#Token	Time	Short	Medium	Long	Overall
0.0	3.6K	36s	**67.6**	59.4	59.1	62.0
0.1	3.4K	30s	67.0	59.7	59.1	61.9
0.2	2.7K	18s	66.0	**60.2**	59.2	61.8
0.3	1.9K	11s	66.4	**60.2**	**59.8**	**62.1**
0.4	0.8K	8s	65.6	58.0	58.3	60.6
0.5	0.3K	7s	63.1	54.9	50.2	56.1
1.0	0.0K	6s	60.3	51.4	44.1	52.0
rnd	1.9K	11s	65.7	55.8	56.0	59.1

Table 4 isolates the role of each source on Video-MME with Long-LLaVA-7B. Without retrieval, simply concatenating auxiliary text helps, but less than query-aware selection. The full non-retrieved combination reaches 59.8%, whereas enabling retrieval raises the same combination to 62.1%. Among single-source retrieved settings, ASR is strongest at 61.6%, showing that spoken content is highly informative in Video-MME. However, the best result still comes from combining all three sources, which indicates complementarity between speech, scene text, and object grounding.

The object branch is particularly useful when the answer depends on counting, locations, or relations rather than narration. This is visible in the Christmas-tree example of Figure 4, where the retrieved detection text explicitly counts apples, candles, and berries, correcting the baseline model's mistaken answer.

5.2 Retrieval Threshold and Efficiency

Table 5 shows the trade-off controlled by the retrieval threshold. Lower thresholds retrieve more text and are slower, but they do not necessarily improve accuracy. The best overall score, 62.1%, occurs at $t = 0.3$ with 1.9K retrieved tokens and 11 seconds average inference time per question. In contrast, $t = 0.0$ retrieves 3.6K tokens and requires 36 seconds while reaching 62.0%. The random retrieval control at the same token budget and latency drops to 59.1%, confirming that the gain comes from query-aware selection rather than from adding arbitrary text.

Figure 3 further shows that Video-RAG is robust to the frame budget. Across 8, 16, 32, 64, 128, and 256 sampled frames, the gain remains large, between 8.7 and 11.4 points. The benefit is strongest at 16 frames and remains above 9 points even at 256 frames, indicating that the method is not merely compensating for extremely sparse visual input. Instead, it consistently improves evidence selection.

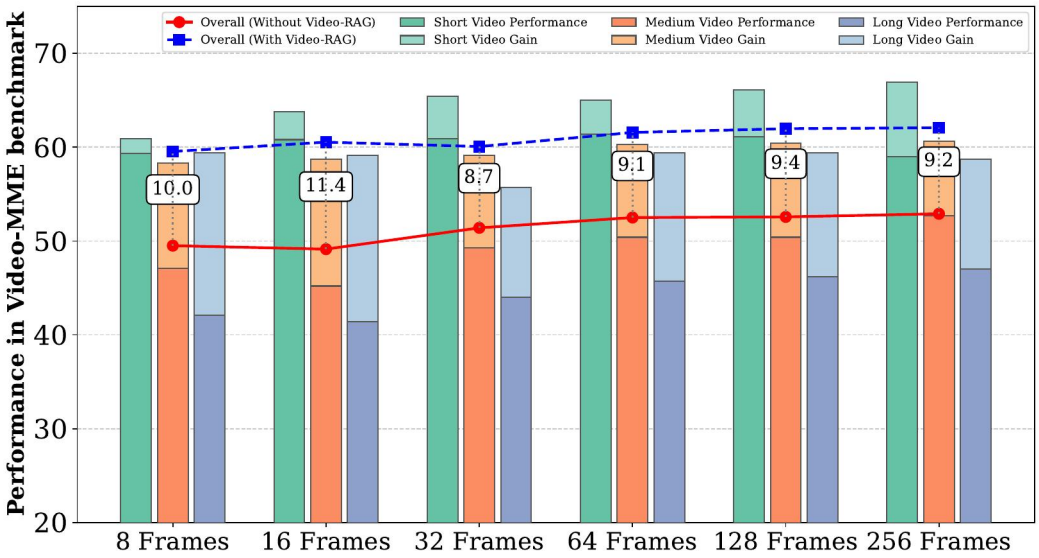

Figure 3: Effect of the frame budget on Video-MME for a 7B long-video LVLM. Video-RAG consistently improves overall accuracy across 8 to 256 sampled frames, with gains ranging from 8.7 to 11.4 points. The largest gain appears at 16 frames, showing that retrieved auxiliary text is especially valuable when the visual budget is tight.

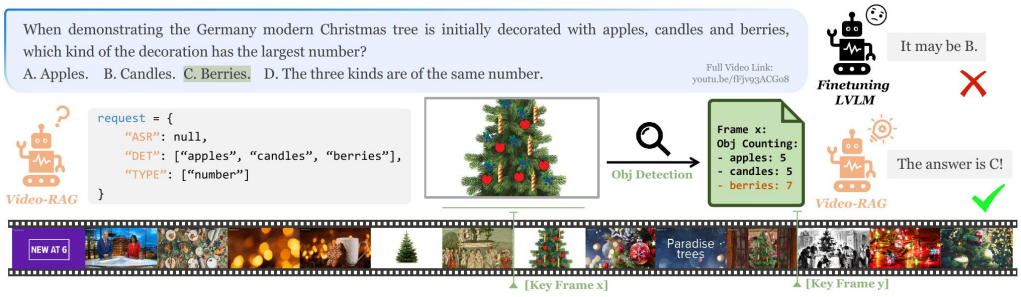

Figure 4: Qualitative example on Video-MME with LLaVA-Video [36]. The question asks which decoration appears most often when a Christmas tree is initially decorated. The baseline LVLM predicts candles, while Video-RAG retrieves an object-count description from relevant keyframes and correctly answers berries.

5.3 Why Does It Help?

Figure 5 provides evidence for the mechanism behind the gains. In the qualitative example, Grad-CAM [21] shifts attention away from query-irrelevant broadcast frames and toward the keyframe containing the decorated tree when auxiliary text is present. The t-SNE visualization [27] also shows that query-text features and keyframe-vision features become more aligned: the center distance decreases from 33.79 without auxiliary text to 24.84 with auxiliary text. While this analysis is illustrative rather than exhaustive, it supports the intended role of Video-RAG as a lightweight alignment layer between the question and visually relevant evidence.

More broadly, the ablations suggest two distinct effects. First, retrieval removes irrelevant auxiliary text, which reduces noise and token cost. Second, object-derived scene-graph descriptions expose discrete facts such as counts and spatial relations in a form the LVLM can use more reliably than raw pixels alone. This is especially useful for long videos, where the relevant visual evidence may appear only briefly.

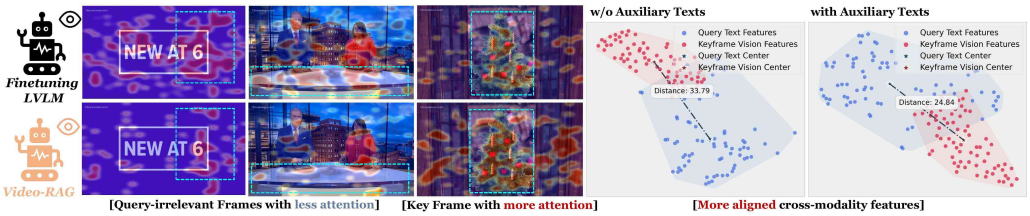

Figure 5: Attention and representation analysis for the example in Figure 4. With auxiliary text, Grad-CAM [21] focuses more strongly on the relevant tree frame and less on distractor frames. In the t-SNE [27] projection, the distance between query-text and keyframe-vision feature centers drops from 33.79 to 24.84, indicating improved cross-modal alignment.

6 Conclusion

Video-RAG is a simple way to improve long-video comprehension without retraining LVLMs. Instead of extending the visual context window, it extracts speech, scene text, and object-level evidence from the video, retrieves the query-relevant subset, and injects that evidence back into the prompt. The method is training-free, uses open-source components throughout, and consistently improves performance on Video-MME, MLVU, and LongVideoBench. The empirical picture is clear: when long-video reasoning is limited by context budget and weak query-frame alignment, a small amount of retrieved auxiliary text can be more useful than additional raw frames.

References

[1] Md Adnan Arefeen, Biplob Debnath, Md Yusuf Sarwar Uddin, and Srimat Chakradhar. irag: An incremental retrieval augmented generation system for videos. *arXiv preprint arXiv:2404.12309*, 2024.

[2] Lin Chen, Xilin Wei, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Bin Lin, Zhenyu Tang, et al. Sharegpt4video: Improving video understanding and generation with better captions. *arXiv preprint arXiv:2406.04325*, 2024.

[3] Jiajun Fei, Dian Li, Zhidong Deng, Zekun Wang, Gang Liu, and Hui Wang. Video-ccam: Enhancing video-language understanding with causal cross-attention masks for short and long videos. *arXiv preprint arXiv:2408.14023*, 2024.

[4] Chaoyou Fu, Yuhan Dai, Yondong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. *arXiv preprint arXiv:2405.21075*, 2024.

[5] Chaoyou Fu, Haojia Lin, Xiong Wang, Yi-Fan Zhang, Yunhang Shen, Xiaoyu Liu, Haoyu Cao, Zuwei Long, Heting Gao, Ke Li, et al. Vita-1.5: Towards gpt-4o level real-time vision and speech interaction. *arXiv preprint arXiv:2501.01957*, 2025.

[6] Gautier Izacard, Mathilde Caron, Lucas Hosseini, Sebastian Riedel, Piotr Bojanowski, Armand Joulin, and Edouard Grave. Unsupervised dense information retrieval with contrastive learning. *arXiv preprint arXiv:2112.09118*, 2021.

[7] JaidedAI. Easyocr. https://github.com/JaidedAI/EasyOCR, 2023.

[8] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with gpus. *IEEE Transactions on Big Data*, 7(3):535–547, 2019.

[9] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33:9459–9474, 2020.

[10] Dongxu Li, Yudong Liu, Haoning Wu, Yue Wang, Zhiqi Shen, Bowen Qu, Xinyao Niu, Guoyin Wang, Bei Chen, and Junnan Li. Aria: An open multimodal native mixture-of-experts model. *arXiv preprint arXiv:2410.05993*, 2024.

[11] KunChang Li, Yinan He, Yi Wang, Yizhuo Li, Wenhai Wang, Ping Luo, Yali Wang, Limin Wang, and Yu Qiao. Videochat: Chat-centric video understanding. *arXiv preprint arXiv:2305.06355*, 2023.

[12] Bin Lin, Bin Zhu, Yang Ye, Munan Ning, Peng Jin, and Li Yuan. Video-llava: Learning united visual representation by alignment before projection. *arXiv preprint arXiv:2311.10122*, 2023.

[13] Qinghong Lin. Vlog: Transform video as a document with chatgpt, clip, blip2, grit, whisper, langchain. https://github.com/showlab/VLog, 2023.

[14] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. Llava-next: Improved reasoning, ocr, and world knowledge. https://llava-vl.github.io/blog/2024-01-30-llava-next/, 2024.

[15] Zuyan Liu, Yuhao Dong, Ziwei Liu, Winston Hu, Jiwen Lu, and Yongming Rao. Oryx mllm: On-demand spatial-temporal understanding at arbitrary resolution. *arXiv preprint arXiv:2409.12961*, 2024.

[16] Juhong Min, Shyamal Buch, Arsha Nagrani, Minsu Cho, and Cordelia Schmid. Morevqa: Exploring modular reasoning models for video question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13235–13245, 2024.

[17] OpenAI. Gpt-4o system card. https://openai.com/index/gpt-4o-system-card/, 2024.

[18] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pages 8748–8763, 2021.

[19] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *International Conference on Machine Learning*, pages 28492–28518, 2023.

[20] Machel Reid, Nikolay Savinov, Denis Teplyashin, Dmitry Lepikhin, Timothy Lillicrap, Jean-Baptiste Alayrac, Radu Soricut, Angeliki Lazaridou, Orhan Firat, Julian Schrittwieser, et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.

[21] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 618–626, 2017.

[22] Yuzhang Shang, Bingxin Xu, Weitai Kang, Mu Cai, Yuheng Li, Zehao Wen, Zhen Dong, Kurt Keutzer, Yong Jae Lee, and Yan Yan. Interpolating video-llms: Toward longer-sequence lmms in a training-free manner. *arXiv preprint arXiv:2409.12963*, 2024.

[23] Yunhang Shen, Chaoyou Fu, Peixian Chen, Mengdan Zhang, Ke Li, Xing Sun, Yunsheng Wu, Shaohui Lin, and Rongrong Ji. Aligning and prompting everything all at once for universal visual perception. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13193–13203, 2024.

[24] Yan Shu, Peitian Zhang, Zheng Liu, Minghao Qin, Junjie Zhou, Tiejun Huang, and Bo Zhao. Video-xl: Extra-long vision language model for hour-scale video understanding. *arXiv preprint arXiv:2409.14485*, 2024.

[25] Enxin Song, Wenhao Chai, Guanhong Wang, Yucheng Zhang, Haoyang Zhou, Feiyang Wu, Xun Guo, Tian Ye, Yan Lu, Jenq-Neng Hwang, et al. Moviechat: From dense token to sparse memory for long video understanding. *arXiv preprint arXiv:2307.16449*, 2023.

[26] Yin Song, Chen Wu, and Eden Duthie. aws-prototyping/long-llava-qwen2-7b. https://huggingface.co/aws-prototyping/long-llava-qwen2-7b, 2024.

[27] Laurens Van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of Machine Learning Research*, 9(11), 2008.

[28] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.

[29] Xiaohan Wang, Yuhui Zhang, Orr Zohar, and Serena Yeung-Levy. Videoagent: Long-form video understanding with large language model as agent. *arXiv preprint arXiv:2403.10517*, 2024.

[30] Haoning Wu, Dongxu Li, Bei Chen, and Junnan Li. Longvideobench: A benchmark for long-context interleaved video-language understanding. *arXiv preprint arXiv:2407.15754*, 2024.

[31] Lin Xu, Yilin Zhao, Daquan Zhou, Zhijie Lin, See Kiong Ng, and Jiashi Feng. Pllava: Parameter-free llava extension from images to videos for video dense captioning. *arXiv preprint arXiv:2404.16994*, 2024.

[32] Ce Zhang, Taixi Lu, Md Mohaiminul Islam, Ziyang Wang, Shoubin Yu, Mohit Bansal, and Gedas Bertasius. A simple llm framework for long-range video question-answering. *arXiv preprint arXiv:2312.17235*, 2023.

[33] Lu Zhang, Tiancheng Zhao, Heting Ying, Yibo Ma, and Kyusong Lee. Omagent: A multi-modal agent framework for complex video understanding with task divide-and-conquer. *arXiv preprint arXiv:2406.16620*, 2024.

[34] Peiyuan Zhang, Kaichen Zhang, Bo Li, Guangtao Zeng, Jingkang Yang, Yuanhan Zhang, Ziyue Wang, Haoran Tan, Chunyuan Li, and Ziwei Liu. Long context transfer from language to vision. *arXiv preprint arXiv:2406.16852*, 2024.

[35] Yuanhan Zhang, Bo Li, Haotian Liu, Yong Jae Lee, Liangke Gui, Di Fu, Jiashi Feng, Ziwei Liu, and Chunyuan Li. Llava-next: A strong zero-shot video understanding model. https://llava-vl.github.io/blog/2024-04-30-llava-next-video/, 2024.

[36] Yuanhan Zhang, Jinming Wu, Wei Li, Bo Li, Zejun Ma, Ziwei Liu, and Chunyuan Li. Video instruction tuning with synthetic data, 2024. URL https://arxiv.org/abs/2410.02713.

[37] Junjie Zhou, Yan Shu, Bo Zhao, Boya Wu, Shitao Xiao, Xi Yang, Yongping Xiong, Bo Zhang, Tiejun Huang, and Zheng Liu. Mlvu: A comprehensive benchmark for multi-task long video understanding. *arXiv preprint arXiv:2406.04264*, 2024.